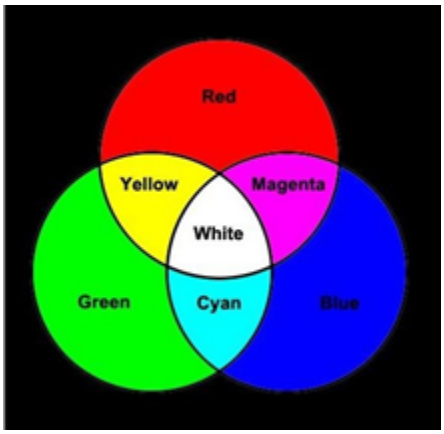
American Computer Science League

2021-2022 ● Contest 1: Fibonacci Clock ● Senior Division

PROBLEM: ACSL's version of Philippe Chretien's "Fibonacci Clock" displays time by changing the colors displayed in 5 squares, whose side lengths correspond to the first 5 Fibonacci numbers (1, 1, 2, 3, and 5). Given the colors of the squares on the clock face, you must output the time that is represented in hh:mm:ss format. The colors will be given to you as a single string of 5 uppercase characters representing the lower 1x1 square, the upper 1×1 square, the 2×2 square, the 3×3 square, and the 5×5 square in that order. (See https://basbrun.com/2015/05/04/fibonacci-clock/.)

Each square has a color: red, green, blue, cyan, magenta, yellow, or white. All red squares represent only the number of hours; green squares represent only minutes; and blue squares represent only seconds. However, intersecting regions (in the Venn Diagram shown) are used to represent 2 different values: yellow represents both hours and minutes; magenta represents both hours and seconds; and cyan represents both minutes and seconds. White squares are ignored.

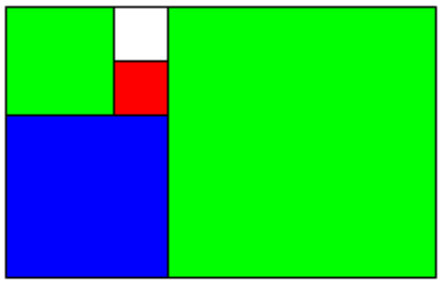


For both minutes and seconds, the sum of the values of the lengths is multiplied by 5 so that the number of minutes and seconds are in intervals of 5 between 0 and 60 inclusive.

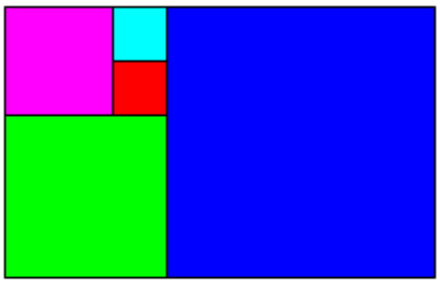
However, if the number of minutes or seconds is 60 or more and/or if the number of hours is 12 or more, change it to a valid time from 00:00:00 to 11:56:55. For example, 09:60:00 would become 10:00:00 and 12:00:60 would be 00:01:00.

In this modified version of the clock, the time displayed on the clock in the picture above is 01:15:35. The hours are represented by the red 1x1 square. The minutes are represented by the green 3×3 square (3*5=15), and the seconds are represented by the blue 2×2 and the blue 5×5 squares (2+5=7 and 7*5=35).

For example, the 2 illustrations below represent the first two sample inputs. The string "RWGBG" represents 1 hour, (2+5) * 5 = 35 minutes, and 3 * 5 = 15 seconds. The string "RCMGB' represents 3 hours (1+2), 20 minutes ((1+3)*5), and 40 seconds ((5+1+2)*5) or 03:20:40.



RWGBG - 01:35:15



RCMGB - 03:20:40

INPUT: There are 5 sets of data. Each set has a single 5-character string of 7 possible uppercase characters, such as "RWGBG". The possible letters are R, G, B, C, M, Y, and W. The order is the lower 1×1, the upper 1×1, the 2×2, the 3×3, and the 5×5 square. We guarantee that the input will represent a valid time from 00:00:00 to 11:56:55.

OUTPUT: For each line of data, print the time in hours, minutes, and seconds formatted as hh:mm:ss.

SAMPLE INPUT:

```
RWGBG
RCMGB
BYYGR
MRGBW
YYYYY
```

SAMPLE OUTPUT:

```
1. 01:35:15
2. 03:20:40
3. 08:30:05
4. 02:10:20
5. 01:00:00
```

TEST DATA

TEST INPUT:

```
WGBGR
CGRMY
MMMMC
CCCYY
WWWWW
```

TEST OUTPUT:

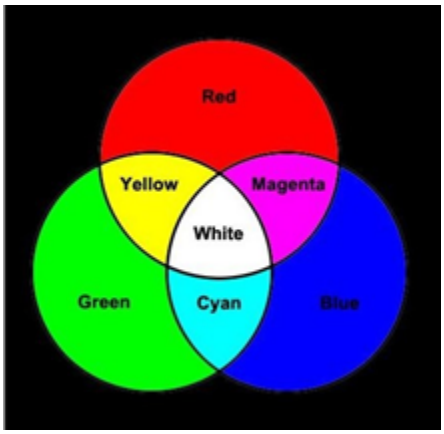
```
1. 05:20:10
2. 10:35:20
3. 07:26:00
4. 09:00:20
5. 00:00:00
```

美国计算机科学联赛

2021-2022 ● 第一轮: 斐波那契时钟 ● 高级组

问题：美国计算机科学联赛版本的菲利普·克雷蒂安“斐波那契时钟”是通过改变由五种正方形呈现的颜色来显示时间的，它们的边长分别对应斐波那契数字的前五个数（1，1，2，3，5）。你要根据钟面上正方形的颜色，以 hh:mm:ss 的格式输出其对应的时间。这些颜色会以由五个大写字母组成的字符串的方式给到你，它们依次代表下方 1x1正方形，上方 1x1正方形，2x2正方形，3x3正方形，以及5x5正方形。 (见 https://basbrun.com/2015/05/04/fibonacci-clock/.)

每个正方形都有一种颜色：红色、绿色、蓝色、青色、紫红色、黄色或白色。 所有红色正方形仅表示小时数； 绿色正方形仅表示分钟数； 蓝色正方形仅表示秒数。但是，交叉区域（如维恩图所示）表示两种不同的值：黄色代表小时和分钟；紫红色代表小时和秒；青色代表分钟和秒。白色正方形可以忽略。

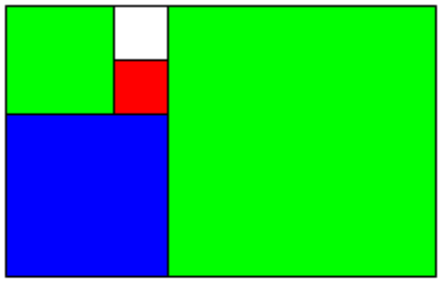


将正方形边长的值相加再乘以 5 得出分钟和秒，这样分钟数和秒数以 5 为间隔，介于 0 到 60 之间。

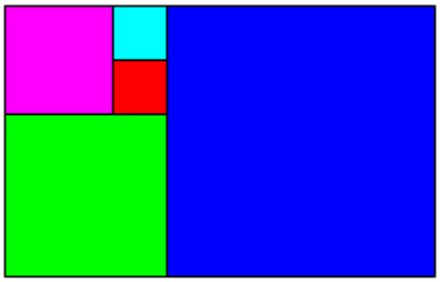
但是，若分钟数或者秒数大于等于 60，并且/或者小时数大于等于12，则需要将其变为 00:00:00 到 11:56:55 之间的一个有效时间。例如，09:60:00要变成 10:00:00，12:00:60 要变成 00:01:00。

在这个修改版的时钟里，最上方时钟所显示的时间是 01:15:35。1x1 的红色正方形表示小时数， 3x3 的绿色正方形 (3*5=15)表示分钟数，2x2 的蓝色正方形和 5x5 的蓝色正方形 (2+5=7 和 7*5=35) 表示秒数。

例如，以下两幅图表示的是前两个示例输入。字符串 “RWGBG” 代表 1 小时， (2+5) * 5 = 35分钟, 3 * 5 = 15秒。字符串 “RCMGB” 表示 3 小时（1+2），20 分钟 ((1+3)*5), 40 秒 ((5+1+2)*5) 或 03:20:40。



RWGBG - 01:35:15



RCMGB - 03:20:40

输入：有五组数据。每组包含一个由 5 个字母组成的字符串，如 “RWGBG”，它们可能来自 7 个大写字母，这些可能的字母是 R, G, B, C, M, Y 和 W。顺序为 1x1 正方形, 上方 1x1 正方形, 2x2 正方形, 3x3 正方形, 以及 5x5 正方形。确保输入内容能表示 00:00:00 到 11:56:55中的一个有效时间。

输出：打印输出每一行数据中由小时、分钟和秒组成的时间，格式为 hh:mm:ss。

示例输入：

```
RWGBG
RCMGB
BYYGR
MRGBW
YYYYY
```

示例输出：

```
1. 01:35:15
2. 03:20:40
3. 08:30:05
4. 02:10:20
5. 01:00:00
```

美国计算机科学联赛

2021-2022 ● **第一轮**: 斐波那契时钟 ● 高级组

测试数据

测试输入:

```
WGBGR
CGRMY
MMMMC
CCCYY
WWWWW
```

测试输出:

```
1. 05:20:10
2. 10:35:20
3. 07:26:00
4. 09:00:20
5. 00:00:00
```